MILITARY SPECIFICATION

FILTER, AIR, DETECTOR (for M229 REFILL KIT)

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification establishes requirements for one type of kit.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-D-51388 - Detector Cell, Chemical Agent.
MIL-D-51456 - Detector Cell, Chemical Agent, Ultrasonically Welded.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-810 - Environmental Test Methods.

FSC 6665

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

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DRAWINGS AND TECHNICAL DATA PACKAGE LIST (TDPL)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

TDPL5-15-8428	-	Filter, Air, Detector.
D136-40-79-2	-	Tester, Agent Conversion and Detection, Q197A1.
D136-41-2136	-	Tester, Leakage and Resistance, Filter Assembly, Q198.
E136-41-1520	-	Chemical Vapor Generator, Q219

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on TDPL5-15-8428 and on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings and in this specification.

3.1.2 Components. All components of this kit shall conform to the specifications and drawings listed on TDPL5-15-8428 and subsidiary drawings.

3.2 Assembly. The kit shall be assembled as specified on Drawing C5-15-8428.

3.3 Performance.

3.3.1 Air filter (Drawing C5-15-5307).

3.3.1.1 Pressure drop across the filters. The pressure drop across the filters housed in the air filter shall be between 1.0 inch (2.5 centimetres) of water and 3.75 inches (9.5 centimetres) of water at an airflow rate through the filters of 1.1 $\pm$ 0.1 liters per minute when tested as specified in 4.4.4.1.1.

3.3.1.2 Leakage. Air leakage through the interface seal between the air filter and air filter insert shall not exceed 3.0 cubic centimetres per 15 seconds at an air pressure difference across the interface seal of 45.0 $\pm$ 5.0 inches (114.3 $\pm$ 12.7 centimetres) of water when tested as specified in 4.4.4.1.2.

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3.3.1.3 Filter conversion of O,S, diethyl methylphosphonothiolate (OSDMP). The air filter when tested under the conditions and parameters listed in table I shall affect a conversion of OSDMP which shall produce an electrical output from the detector cell of at least 15.0 millivolts above background with a CT of 0.30 ± 0.10 gamma-minutes per liter when tested in accordance with 4.4.4.1.3.

3.3.1.4 Moisture content. The moisture content of the air filter shall be from 1.0 milligram (mg) to 4.0 mg of water when tested as specified in 4.4.4.4.4.

Table I. Conditions and parameters

Condition	Parameter
Filter conditioning	*
OSDMP concentration	0.30 ± .10 gamma/l
Flowrate	1.1 ℓ/min
Temperature	92.5° ± 2.5° F (33.6° ± 1.4° C)
**Cell sensitivity	21 to 36 mv

*Prior to testing, the air filter shall be aerated for at least 15 minutes but not more than 2.0 hours.

**Cells used for the filter conversion tests shall initially conform to MIL-D-51388 or MIL-D-51456 and shall be requalified to the background, noise, and drift and sensitivity requirements of MIL-D-51388 at least once every three months. Noise and background shall be recorded for a period of 10 and 2 minutes respectively. After cell has been in operation for at least 10 minutes, cell background shall not exceed 30 mv during any 12 hour continuous test period.

3.3.2 Low temperature storage. After being stored at -60° ± 5° F (-51.1° ± 2.8° C) for at least 14 days, and then at 80° ± 5° F (26.7° ± 2.8° C) for a minimum of 12 hours as specified in 4.3.3.1, the air filter in the kit shall be capable of satisfying the conversion requirements of 3.3.1.3.

3.3.3 High temperature storage. After being stored at 150° ± 5° F (65.6° ± 2.8° C) for at least 14 days, and then allowed to return to room temperature as specified in 4.3.3.2, the air filter in the kit shall be capable of satisfying the conversion requirement of 3.3.1.3.

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3.3.4 Transportation vibration. After being subjected to the vibration requirements of MIL-STD 810, method 514, procedure XI, part 2, as specified in 4.3.3.3., the air filter in the kit shall be capable of satisfying the moisture content requirement of 3.3.1.4 and the conversion requirement of 3.3.1.3.

3.4 Preproduction. Prior to the start of regular production, a preproduction sample of kit shall be produced in accordance with this specification for examination and test (see 4.3).

3.5 Workmanship. The kit shall be free from damage and evidence of leakage.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections.

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction

4.3.1 Sample. A preproduction sample of 12 kits shall be produced using the same methods, materials, equipment, and processes as will be used in regular production. Air filters used in the production of the sample of kits shall have been examined and tested in accordance with 4.4.

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4.3.2 Inspection.

4.3.2.1 For examination and environmental test. The preproduction sample of kits shall be examined in accordance with the classification of defects (4.4.3.2(a) and then divided into groups and tested as specified in table II.

4.3.3 Tests. Tests shall be conducted as follows:

Table II. Preproduction sample inspection

Quantity of kits	Component	Requirement	Component sample size	Acc/rej	Test paragraph
4	Air filters	Low temperature storage 3.3.2	55	Acc 2, rej 3	4.3.3.1
4	Air filters	High temperature storage 3.3.3	55	Acc 9, rej 10	4.3.3.2
4	Air filters	Transportation vibration 3.3.4			4.3.3.3
		converstion	55	Acc 2, rej 3	
		moisture content	44	Acc 0, rej 1	

4.3.3.1 Low temperature storage. Examine each list and its contents to establish the pretest condition of the kit. Reassemble the kit. Do not band, package, or pack the kit. Place the specified quantity (see table II) of kits in a test chamber already at the temperature specified in 3.3.2. Maintain the test chamber at the specified temperature required by 3.3.2 for at least 14 days. Then, increase the temperature of the test chamber to 80° ± 5° F (26.7° ± 2.8° C). Maintain the test chamber at this temperature at least 12 hours. Remove the sample of kits from the test chamber. Subject the air filter sample required by table II to the conversion test required by 3.3.1.3. Evaluate the findings of the post storage test against the requirement of 3.3.2.

4.3.3.2 High temperature storage. Examine each kit and its contents to establish the pretest condition of the kit. Reassemble kit. Do not band, package, or pack the kit. Place the specified quantity (see table II) of kits in a test chamber already at the specified temperature required by 3.3.3. Maintain the test chamber at the temperature required by 3.3.3 for at least 14 days. Then, remove the refill kit sample from the test chamber. Open the kits, and allow the kit contents to cool to room temperature. Subject the air filter sample required by table II to the conversion test required by 3.3.1.3. Evaluate the findings of the post storage test against the requirements of 3.3.3.

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4.3.3.3 Transportation vibration. Examine each kit and its contents to establish the pretest condition of the kit. Reassemble the kit. Subject the specified quantity (see table II) of kits to the vibration test described in MIL-STD-810, method 514, procedure XI, part 2. After completion of the prescribed test, subject the air filter sample required by table II to the conversion and moisture tests required by 3.3.1.3 and 3.3.1.4. Evaluate the findings of the post vibration tests against the requirements of 3.3.4.

4.3.4 Acceptance/rejection criteria. The preproduction sample shall comply with the examination and tests specified in 4.3.3 and 4.4.4 to be acceptable. The supplier shall obtain written approval from the contracting officer prior o proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting.

(a) Kits. A lot shall consist of the kits offered for acceptance at one time which have been produced by one manufacturer, at one plant, under essentially the same manufacturing conditions, and in essentially the same time frame. However, not more than one lot of air filters shall be represented in any one lot of finished refill kits.

(b) Air filters. A lot shall consist of the air filters manufactured from the filter material which has been impregnated from one batch of silver nitrate solution.

4.4.2 Sampling.

4.4.2.1 For examination. Sampling of kits or air filters shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For test. Sampling of air filters shall be conducted in accordance with table III.

4.4.3 Inspection procedure.

4.4.3.1 For examination and tests. Sample kits and air filters shall be examined and tested in accordance with the classification of defects (4.4.3.2), and MIL-STD-105, and table III.

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Table III. Air filter sampling and inspection

Requirement	Military standard	AQL (%)	Test paragraph
Pressure drop 3.3.1.1	MIL-STD-105	2.5	4.4.4.1.1
Leakage 3.3.1.2	MIL-STD-105	2.5	4.4.4.1.2
OSDMP conversion 3.3.1.3	MIL-STD-105	2.5	4.4.4.1.3
Moisture content 3.3.1.4	MIL-STD-105	*2.5	4.4.4.1.4
*Inspection level S-4			

4.4.3.2 Classification of defects.

(a) Filter, air, detector (Drawing C5-15-8428).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Component missing, incorrect or incorrectly assembled
102	Printed instructions missing, incorrect, or illegible
103	Damage: kit ripped, torn, or punctured, workmanship (3.5)

(b) Package assembly (Drawing C5-15-5309).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Air filter package: inner or outer package missing, incorrect, or incorrectly assembled
102	Air filter package: inner or outer package ripped, torn, or punctured
103	Air filter component or filter missing, incorrect, or incorrectly assembled

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104	Air filter or insert cracked, gouged or warped
105	Filters ripped, torn, punctured, or cracked

(c) Preparation for delivery (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Packaging or packing component missing, incorrect, or incorrectly assembled
102	Packaging or packing component damaged
103	Markings missing, incorrect, or illegible

4.4.4 Tests. The following tests shall be conducted at an ambient temperature of not less than 70° F (21.1° C).

4.4.4.1 Air filter.

4.4.4.1.1 Pressure drop across the filters. Determine the pressure drop utilizing the Q198(D 136-41-2136) in accordance with the instruction manual.

4.4.4.1.2 Leakage. Determine the leakage utilizing the Q198(D 136-41-2136) in accordance with the instruction manual.

4.4.4.1.3 OSDMP conversion. Utilizing the Q197A1(DL 136-40-79-2) in accordance with the instruction manual, condition the air filter for the required period of time. After completion of the conditioning period, the filter shall be tested within 30 minutes. Determine the conversion capabilities of the air filter, utilizing the Q197A1 or Q219(E136-41-1520) test equipment in accordance with the instruction manual. Prior to conducting conversion test to the established concentrations of OSDMP, (see 6.9) a determination of the output signal of the unfiltered effluent from the Q219 generator shall be made. The output of the Q197A1 shall be no greater than 15.0 millivolts in 3.0 minutes when tested in accordance with the Q197A1 instruction manual but without an air filter installed. The concentration of OSDMP coming from the Q219 shall be determined prior to running any test and every 4 hours during testing and on completion of testing.

4.4.4.1.4 Moisture content. Remove the air filter from its package and weigh the air filter on an analytical balance with an accuracy of 10^{-4} grams. Place the air filter in a forced air oven preconditioned at

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67° ± 5° centigrade (152.6° ± 9° F) and dry the air filter for one hour. Remove the air filter and place it in a desiccator containing concentrated sulfuric acid and allow the air filter to cool for at least 15 minutes. Remove the air filter from the desiccator and weigh immediately. The difference in weight shall not exceed the specified amount.

5. PACKAGING

5.1 Preservation, packaging, and packing, level A. The kit shall be preserved, packaged, and packed in accordance with the specifications and drawings listed on C5-15-8428 and subsidiary drawings.

5.2 Marking. In addition to any special marking required by the contract, unit packages, shall be marked in accordance with MIL-STD-129. Exterior shipping containers shall be marked in accordance with Drawing C5-15-8428.

6. NOTES

6.1 Intended use. This specification establishes requirements for the kit of expendable air filters to extent the life of the M229 refill kit in the field.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preproduction:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

6.3 Manufacture of conversion filters (filter, top; Drawing B5-15-4723). Edgewood Arsenal Directive No. 644 is a suggested method of manufacturing conversion filters. However, this method is not mandatory for use and in no way relieves the contractor from performing the inspections as set forth herein.

6.4 Storage life limitation. The storage life of the kit is two years from the date of manufacture.

6.5 International interest. Certain provisions of this specification are the subject of international standardization agreement QSTAG 251. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned,

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the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

6.6 Component specifications. The component specifications for this item are as follows:

MIL-D-51388(EA) - Detector Cell, Chemical Agent
MIL-D-51399(EA) - O,S - Diethyl Methylphosphonothiolate
MIL-D-51456(EA) - Detector Cell, Chemical Agent, Ultrasonically Welded.

Custodians:

Army - EA

Preparing activity:

Army - EA

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